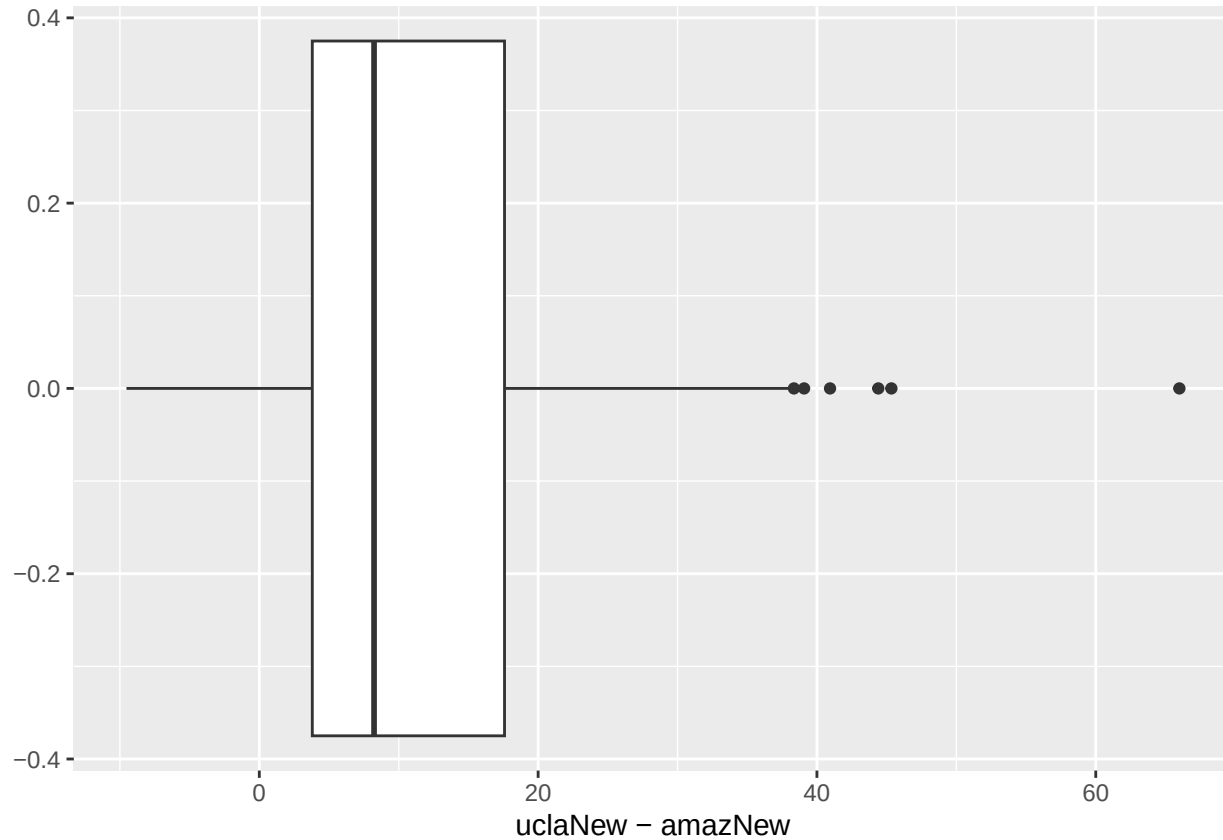


```
books <- read.csv("https://raw.githubusercontent.com/njlytal/MATH314/main/books.csv")
```

```
suppressMessages(library(ggplot2))
ggplot(
  data = books,
  aes(uclaNew - amazNew)
) +
  geom_boxplot()
```



```
diff <- books$uclaNew - books$amazNew
xbar <- mean(diff)
n <- length(diff)
df <- n - 1
alpha <- 0.05
```

```
tstar <- qt(1 - alpha / 2, n - 1) # critical value
stderr <- sd(diff) / sqrt(n) # sd(diff) == sample stdev
margin_err <- tstar * stderr
ci <- c(xbar - margin_err, xbar + margin_err)
```

Hypothesis Test (t-test), 2-sided

```
t <- (mean(diff) - 0) / stderr
(p_value <- 2 * (1 - pt(abs(t), n - 1)))
```

```
## [1] 6.92757e-11
```